

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

School of Information and Communication Technology



SOICT

Project Report

Topic: Sentiment Analysis of Restaurant Reviews

Subject: Web Mining – IT4868E

Supervisor: Professor. Nguyễn Kiêm Hiếu

Group Members:	
Tạ Hồ Thành Đạt	20225482
Trịnh Huỳnh Sơn	20225526
Trần Nhật Minh	20225511
Trần Nam Tuấn Vượng	20225540
Trần Quang Huy	20225500

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Project Role

Member	StudentID	Role
Tạ Hồ Thành Đạt	20225482	Support vector machine method
Trịnh Huỳnh Sơn	20225526	Maximum entropy method
Trần Nhật Minh	20225511	Evaluation and report
Trần Nam Tuấn Vượng	20225540	Data scraping
Trần Quang Huy	20225500	Naive Bayes method

GitHub Repository

[VuongTNT/](#)

[GoogleMap-Review-Scraping-and-Analysis:](#)

[Scrape Google Maps reviews for businesses and perform sentiment analysis to gain insights into customer feedback and trends.](#)



Datasets

[GMap raw data - Google Drive](#)



Part I: Introduction

In this project, we tackle the classic NLP problem of sentiment analysis by focusing on predicting restaurant ratings from user-generated reviews. We collect and scrape new restaurant reviews from Google Maps and aim to classify each review into one of five rating categories (1, 2, 3, 4, or 5), corresponding to the level of customer satisfaction expressed in the text. This task can be viewed as a fine-grained sentiment analysis problem, where subtle differences in opinion, tone, and wording play an important role in determining the final rating.

To address this problem, we formulate it as a multi-class classification task. We explore two main classification methods, namely Support Vector Machines (SVM) and Maximum Entropy (ME), which are well-suited for handling high-dimensional textual features, and compare those to Naive Bayes (NB) as a baseline. By evaluating and comparing these approaches on the newly scraped dataset, we aim to better understand their effectiveness in predicting restaurant ratings from real-world data.

Part II: Dataset

1. Data Collection

We constructed a dataset of almost one million different reviews of restaurants randomly selected within the country of the UK, the US and Vietnam. To do that, we divided this process into two parts based on the tools that we used: location gathering, and review gathering.

First, for location gathering, we used an open-source Google Maps Scraper, to retrieve the information of restaurants. With the provided input coordinates and keywords (in our case, the keywords are “restaurant”, “bar”, “cafe”, etc.), this tool will search for locations within a radius from that coordinates that fit the keywords and return their metadata, including the data ids and review counts which will be fed into the next step.

Next, using the obtained identifiers, we implemented a Python-based data collection pipeline with the requests library to fetch user reviews associated with each restaurant that had more and 10 reviews. By sending HTTP requests to the Google Maps review endpoints, we were able to extract review texts along with their corresponding ratings. This approach enabled us to collect a large number of real-world, user-generated reviews, which were then cleaned and organized for use in our sentiment classification experiments.

Most reviews are written in English, however there are also many non-English ones, especially for that of Vietnamese locations. For reviews in foreign languages, we replaced those with the translated versions into English provided by the HTTP responses. The result is a dataset containing 946 304 unique reviews, each with a corresponding rating. The lengths of these reviews vary widely, from roughly 1 to 837 words (average 28), or 10 to 4096 characters (average 160), including punctuation, spaces and emojis.

The following tables shows examples of reviews with varied lengths in the dataset:

Review	Rate	Word count
In a word, rubbish!	1	4
Not so good service	2	4
Nice place	3	2
Tapas is one of my favourite meals however i was left feeling disappointed by the portion sizes, lack of flavour. I can only base my ratings on previous encounters with this cuisine. Also i found the staff more concerned with gazing into the computer screen rather than attention to the customer. No once was we asked is everything ok with your meal. I left feeling hungry.	3	65
Amazing service great food,atmosphere and people	4	6
Great little country pub. Excellent home cooked food at reasonable prices, good choice of ales, beers and ciders and friendly staff. Highly recommended 👍	5	24

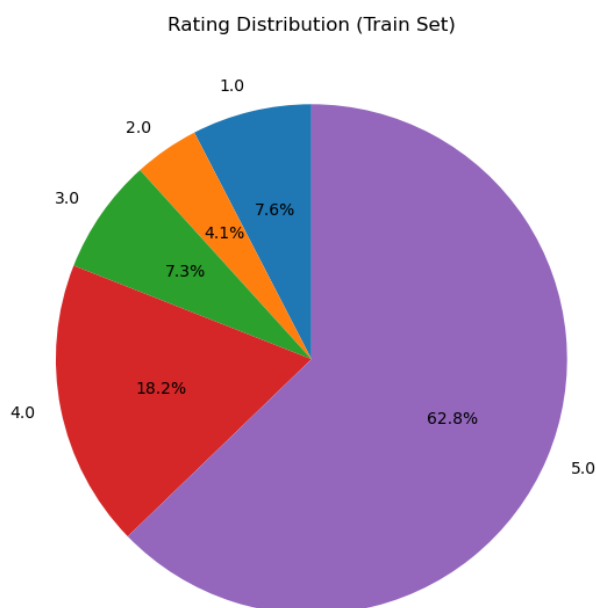
Table 1. Example of small to medium length reviews

Review	Rate	Word count
Food not cooked chicken raw took one bite and had to spit it out looked at it and could see it was flash grilled outside but raw inside. Wrap was just rolled together so as tried to lift it out of pack the lot fell out in the box so had to remake myself with a piece of tomato looked like from a cherry tomato 2 pieces of lettuce 1cm by 5cm and 2 bagons same size and enough sauce to fill a teaspoon absolutely no value	1	135

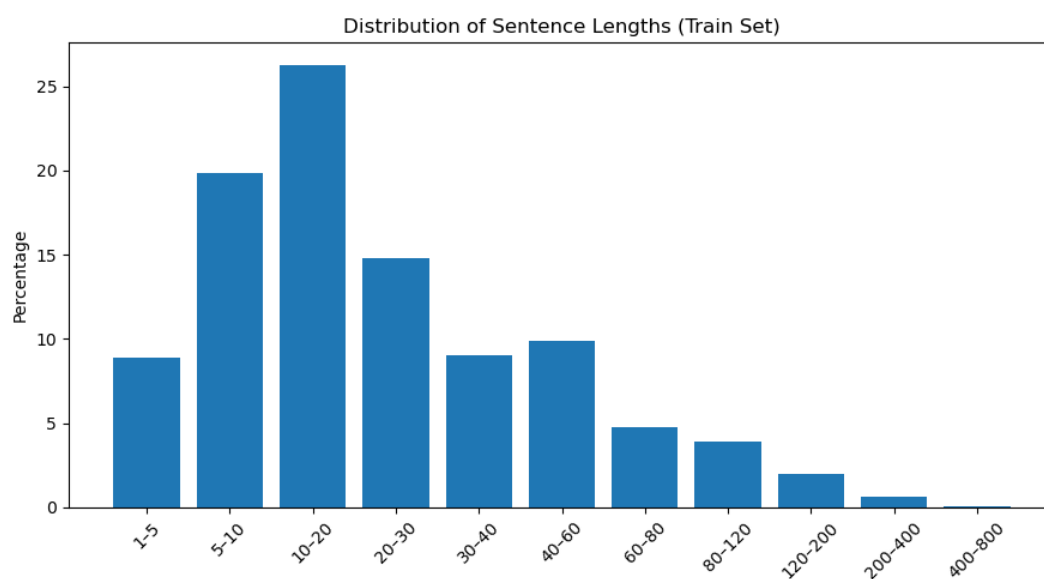
worst food ever received. Asked for some sauce told you have to pay even though those eating in get free and in drive thru get free. Very surly staff didn't care 3 stood infront of me ignoring me when i asked for a bit of sauce. Worst ever		
If I could give 12 stars, I would. Nick & Norman's in Senoia is just fantastic. We used to live in Senoia when they filmed the walking dead and used to frequent the restaurant on a regular basis. We would see the actors in here occasionally but didn't pay much attention because who wants to be bothered trying to eat right. One time, we got to have a good conversation with Norman Redus and Jeffrey Dean Morgan to the point where we actually asked them to sign our Harley Davidson. They were happy to do it. I bet they still remember, lol. Now, a few years later, living in a different state, whenever we get back to Senoia, we immediately come here for dinner. The food is perfect every time and delicious. We recently came to visit for a family wedding and went here first, and of course they didn't disappoint. I think we've been here a few dozen times and never a complaint. Its atmosphere is reminiscent of a speak-easy from a different time, warm and cozy. Go here when you're in town. You will remember and want to go back, and not just for the film status, but the place and the great food. We plan to move back to Senoia in the near future, and this will be the first celebration destination on arrival and many more to come. Good job guys!!	5	235

Table 2. Example of large reviews

The dataset is split into train/val/test sets with ratio 80%:10%:10%. The following graphs show the distributions of the train set:



The distribution of the rating labels in the train set



The distribution of the lengths of sentences in the train set

2.Data preprocessing

Using the nltk library, we apply a processing procedure which contains multiple natural language processing techniques with the primary aim to remove noise from the original data. Some preprocessing techniques were applied in models and here are the following:

- +) *Punctuation removal and lowercasing*: All text is converted to lowercase to maintain uniformity. Punctuation is removed to focus on the meaningful words.
- +) *Stop words removal*: Typically, stop words like "the", "is", "in", etc., are removed as they don't add significant meaning to the text.
- +) *Tokenization*: Break down the text sequence into small tokens for easier processing.
- +) *Lemmatization*: Convert words to their base form, ensuring that words like "running" and "runner" are treated as "run".

The following is an example of how a review is transformed:

Original Text	Great little country pub. Excellent home cooked food at reasonable prices, good choice of ales, beers and ciders and friendly staff. Highly recommended 👍
Punctuation removal and lowercasing	great little country pub excellent home cooked food at reasonable prices good choice of ales beers and ciders and friendly staff highly recommended
Tokenization	['great', 'little', 'country', 'pub', 'excellent', 'home', 'cooked', 'food', 'reasonable', 'prices', 'good', 'choice', 'ales', 'beers', 'ciders', 'friendly', 'staff', 'highly', 'recommended']
Lemmatization	['great', 'little', 'country', 'pub', 'excellent', 'home', 'cooked', 'food', 'reasonable', 'price', 'good', 'choice', 'ale', 'beer', 'cider', 'friendly', 'staff', 'highly', 'recommended']

Part III: Methodology

With the goal of classifying text sequences into five classes, we selected Support Vector Machines (SVM) and Maximum Entropy (ME) as our method, with Naive Bayes (NB) as a baseline.

1. Naïve Bayes (NB)

1.1. Introduction

This model is integrated into a Scikit-learn Pipeline that first transforms the raw, cleaned text into a numerical format using TfidfVectorizer.

- TfidfVectorizer: This component converts text documents into a matrix of TF-IDF features (Term Frequency-Inverse Document Frequency), which helps the model understand the importance of words relative to the entire dataset.
- MultinomialNB: This is a probabilistic learning algorithm commonly used in text classification. It estimates the probability of a specific rating (e.g., Rating 1, Rating 2) based on the frequency of the words present in the review.

Loss and Objective Function

The model operates based on the following mathematical and optimization principles:

- Probabilistic Objective: The Multinomial Naive Bayes model works by calculating the Maximum A Posteriori (MAP) probability to determine which class (rating) a text most likely belongs to.
- Optimization Objective: The script uses GridSearchCV, which aims to maximize the accuracy score. Specifically, it performs a 5-fold cross-validation (cv=5) to find the combination of parameters that yields the highest average best_score_ during training.

1.2. Parameters

These results represent the optimal configuration for the model, discovered by the GridSearchCV process after testing various combinations of settings. GridSearchCV performed a 5-fold cross-validation (cv=5) to determine which parameters yielded the highest average accuracy.

- nb__alpha: 0.1
 - **Definition:** This is the smoothing parameter for the MultinomialNB (Naive Bayes) classifier.
 - **Explanation:** In Naive Bayes, smoothing is used to handle words that might not appear in the training data but appear in the test data, preventing the model from assigning a "zero" probability. Out of the tested values (0.1, 0.5, 1.0), 0.1 was the most effective. This suggests that a lower level of smoothing was sufficient to maintain the model's predictive power without over-correcting the probabilities.
- tfidf__max_df: 0.5
 - **Definition:** This stands for "Maximum Document Frequency" for the TfidfVectorizer.
 - **Explanation:** A value of 0.5 means the model ignores any words that appear in more than 50% of the review documents. These words are often considered "too common" to be useful for distinguishing between different ratings (for example, words like "the" or context-specific common words like "item" or "product"). By filtering these out, the model focuses on more descriptive, high-impact words.
- tfidf__ngram_range: (1, 2)
 - **Definition:** This defines the range of n-grams (sequences of words) the vectorizer extracts.
 - **Explanation:** The result (1, 2) indicates that the model performed best when it looked at both unigrams (single words) and bigrams (pairs of two consecutive words).

2. Maximum Entropy (ME)

2.1. Introduction

ME is implemented as multinomial logistic regression, which directly models the conditional probability of each class given an input document. The method is widely applied in web mining due to its simplicity, efficiency, and strong performance on high-dimensional text data.

2.2. Parameters

The model is trained using TF-IDF features (vocabulary size: 30,000, n-gram range: 1–3) and optimized with the **SAGA solver** for up to **500 iterations**.

Given an input feature vector x , the ME model estimates class probabilities using the softmax function:

$$P(x) = \frac{\exp(w_x x)}{\sum_k \exp(w_k x)}$$

In this formulation:

- x denotes the TF-IDF feature vector of a document.
- w_y is the weight vector corresponding to class y .
- k indexes all sentiment classes (Negative, Neutral, Positive).
- $w_y^\top x$ represents the logit score measuring the relevance between document features and class y .

Model parameters are learned by minimizing the negative log-likelihood (cross-entropy loss) with L2 regularization:

$$L = - \sum_{i=1}^N \log \log P(y_i | x_i) + \lambda \sum_y \|w_y\|^2$$

where:

- N is the number of training samples.
- x_i, Y_i is the feature-label pair of the i -th document.
- λ controls the regularization strength, preventing overfitting by penalizing large weights.

To handle class imbalance in the dataset, **class-weighted loss** is applied, assigning higher importance to the Neutral class during training (1.0 for Negative class, 2.5 for Neutral class and 0.6 for Positive class).

3. Support Vector Machine

3.1. Introduction

SVM, short for Support Vector Machines, are supervised models that learn a hyperplane that best divides the input space into two separate classes by maximizing the margin, distance from the hyperplane to the two closest data points or support vectors. SVM can be used for both classification and regression. We included SVM in our project for its efficiency in text classification tasks specifically.

3.2. Parameters

Our SVM model is trained using TF-IDF features using scikit-learn `TfidfVectorizer`. Multiclass classification is performed via one-vs-one scheme, as it is the default scheme of scikit-learn SVC model. Due to the large amount of training data, we use linear kernel for a more reasonable training time.

Other hyperparameters include:

- `vocab_size = 0.000` : Vocabulary size
- `max_df = 0.5` : Max document frequency
- `ngram_range = (1, 2)` : Range of Ngrams

Part IV: Performance Evaluation

1. Evaluation Metrics

We consider 4 popular metrics: accuracy, precision, recall and F1-score.

- *TP* (True Positive): The prediction is correct and the actual value is positive.
- *FP* (False Positive): The prediction is wrong and the actual value is positive.
- *TN* (True Negative): The prediction is correct and the actual value is negative.
- *FN* (False Negative): The prediction is wrong and the actual value is negative.
- *Accuracy*: Measures the proportion of correct predictions made by the model.

$$Accuracy = \frac{(TP + TN)}{(TP + FP + TN + FN)}$$

- *Precision*: Measures the proportion of true positive predictions among all positive predictions made by the model.

$$Precision = \frac{TP}{(TP + FP)}$$

- *Recall*: Measures the proportion of true positive predictions among all actual positive instances.

$$Recall = \frac{TP}{(TP + FN)}$$

- *F1-Score*: Measures the harmonic mean of precision and recall.

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

In the context of our project, we target to gain the highest score in 2 metrics Accuracy and F1 Score. Accuracy not only provides a straightforward measure of the model's performance, gives a clear picture of how well the model is performing overall but also metric for model's benchmark comparisons. F1-Score is used to ensure all classes are appropriately considered, even if they have different representation sizes. It also provides a deeper insight into the model's ability to correctly classify minority and majority classes alike. By targeting high performance in both metrics, we ensure that the model is not only accurate but also fair and effective in handling data.

2. Results and Comparisons

The following table shows the final results of our approaches. Overall, the SVM model performs the best on all metrics.

Models	Accuracy	Precision	Recall	F1
NB	0.7018	0.6345	0.7018	0.6337
ME	0.7065	0.5545	0.7065	0.6210
SVM	0.7211	0.6698	0.7211	0.6749